RDC Ble Authentication DLL V1.0

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Validation			
Name	Date	Validation (O/N)	Comments

Associated files	
Name	Description

SUMMARY

1 Introduction

This document presents the functionality and how to use the RdcBle DLL for BLE authentication on a RDCB. Once the authentication is successful the RDC start the SPP connection.

2 Dependencies

This DLL was developed and tested with the Windows 10 64 bits operating system, there is no external dependencies.
The BLE module is activated only in active mode on the RDCB.
The product must be first paired with the Windows operating system. If the pairing is done with the system Bluetooth manager at least 2 attempt must be made as the Bluetooth manager badly manage BLE devices with a long advertising period (5s by default for the RDC).

3 Functions

3.1 RdcBleVersion()

3.1.1 Arguments

- uint8_t *pcMajor: unsigned byte pointer to store the major version number
- uint8_t *pcMinor: unsigned byte pointer to store the minor version number

3.1.2 Return value

None.

3.1.3 Description

Get the software version of the DLL.

3.2 RdcBleConnect()

3.2.1 Arguments

- uint32_t RdcId:
  RDC ID. For example if the BLE name is RDC_BLE_57AE4522 then RdcId = 0x57AE4522
- uint8_t *AesKey:
  A pointer to the 16 bytes array of the SPP AES key.
- uint8_t *BtMacAddr:
  A pointer to the 6 bytes array of the PC Bluetooth MAC address. If BtMacAddr is NULL then the DLL will use the first Bluetooth adapter MAC address of the PC. This address is used by the RDC to bond the SPP link with the PC.

3.2.2 Return value

CF §4.

3.2.3 Description

There are 3 main steps in the process:

- Search for the RDC device in all the paired devices.
- Wake up the RDC and establish the BLE connection.
- Perform the authentication procedure.

If the authentication is successful the RDC will try to establish a Bluetooth connection with the MAC address provided during the pairing process. At the first connection with the PC a popup window will appear to accept the connection.

4 Error Codes

4.1 RDCBLE_ERROR_NONE: 0

Value returned when the authentication was successful.

4.2 RDCBLE_ERROR_MEMORY: 3

Value returned if a memory allocation issue append.

4.3 RDCBLE_ERROR_INTERNAL: 4

Value returned if an internal error append.

4.4 RDCBLE_ERROR_VERSION: 5

Value returned if the BLE firmware version is not compatible (not yet used).

4.5 RDCBLE_ERROR_RDC_NOT_PAIRED: 6

Value returned if the targeted RDC is not paired with the Windows OS.

4.6 RDCBLE_ERROR_RDC_CONNECT_FAILED: 7

Value returned if the connection with the RDC failed. Possible causes:

- The RDC is in a wrong mode (storage for example)
- The RDC is out of range.
- The RDC is already connected to another device (PC or phone for example).

4.7 RDCBLE_ERROR_AUTHENTICATION_FAILED: 8

Value returned if the authentication failed. Check that the SPP Key value match the one used in the RDC configuration.